Talha Khan

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SUMMARY

Electrical Engineering student with experience leading and contributing to multidisciplinary engineering projects, including an autonomous vehicle system. Strong background in electrical integration, CAD, and hands-on fabrication, with leadership roles in student organizations and robotics teams. Interested in applying these skills in a collaborative engineering environment.

EDUCATION

University of Michigan — **Dearborn, MI**
BSE Electrical Engineering | ***GPA: 3.87*** — *2024 – 2027(Expected)*

Wayne County Community College — **Belleville, MI**
Associates Of Science | ***GPA: 3.95*** — *2021 – 2024*

TECHNICAL SKILLS

Programming: Python, C, JavaScript, HTML & CSS
Tools: Fusion 360, AutoCAD, MakerBot, Linux, MySQL, MS Office
Technical: Wire Harnessing, Electrical Integration, Autonomous Systems
Soft Skills: Leadership, Collaboration, Mentorship, Communication

EXPERIENCE

Chief Engineer & Treasurer
Intelligent Systems Club — Project Phoenix, UM–Dearborn — *Aug 2025 – Present*

- Led the development and integration of electrical, mechanical, and software systems for an autonomous kart, achieving 3rd place at the Autonomous Karting Series (AKS) National Grand Prix.
- Manage a $30,000 project budget, overseeing procurement and ensuring the project remains on track and under budget toward completion.
- Coordinate design reviews and present technical and financial progress updates to club members and alumni.

Mentor
FIRST Inspires — Team 6190, Canton, MI — *Sept 2021 – May 2026*

- Mentor a 20+ member team in robot design, fabrication, and control system development.
- Lead workshops on CAD, wiring, and mechanical design to build student technical skills.
- Encourage teamwork and structured problem-solving in a competitive robotics environment.

Teaching Assistant (Grader)
University of Michigan–Dearborn — ECE 270, ECE 273, ECE 3731 — *Jan 2025 – Present*

- Assist in grading assignments and exams for multiple electrical and computer engineering courses.
- Provide feedback to reinforce correctness in circuit analysis, embedded systems, and programming concepts.

PROJECTS

Project Phoenix — Autonomous Go-Kart — **2024 – Present**

- Contribute to the design and development of an autonomous go-kart for the Autonomous Karting Series (AKS).
- Fabricated and assembled the vehicle's electrical wire harness, ensuring reliable routing, secure terminations, and safety compliance.
- Integrated and tested sensors, controllers, and power systems to support overall system reliability.
- Interpreted wiring schematics and collaborated across mechanical and software teams for system integration.

Autonomous Mobile Robot (ENGR 100) — **Fall 2024**

- Led a team through the full engineering design process from concept and documentation to functional prototype.
- Designed structural components in CAD and coordinated fabrication and testing.
- Delivered PDR and CDR presentations communicating design decisions and system performance.

Adaptive Light System (ECE 210 — Circuits) — **Fall 2025**

- Designed and implemented an automatic lighting system using a CdS photoresistor and op-amp comparator circuit.
- Developed a voltage divider control system to adjust LED output based on ambient light levels.
- Achieved an estimated 30–35% reduction in energy consumption compared to always-on lighting.
- Built and tested a low-cost energy-efficient system for smart lighting applications.

AWARDS & CERTIFICATIONS

Dean's List | Mentor Collective Certification (Diversity, Equality & Inclusion) | Emerging Scholar

STUDENT ASSOCIATIONS & CLUBS

Intelligent Systems Club | IEEE | Phi Theta Kappa | MSA | Tau Beta Pi